SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Concept Mapping

COURSE NAME: Data Structures **COURSE CODE: CSA0308**

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COURSE OUTCOME COVERED

CO1: *Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.* (BL2, BL3)

Assessment Tool Weightage: 35%

Questions:2 **EACH QUESTIONS CARRIES:** 10 MARKS **Total Marks:**20 **Duration:** ONE HOUR

Q1 A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance.

Q2 A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Code of Conduct:

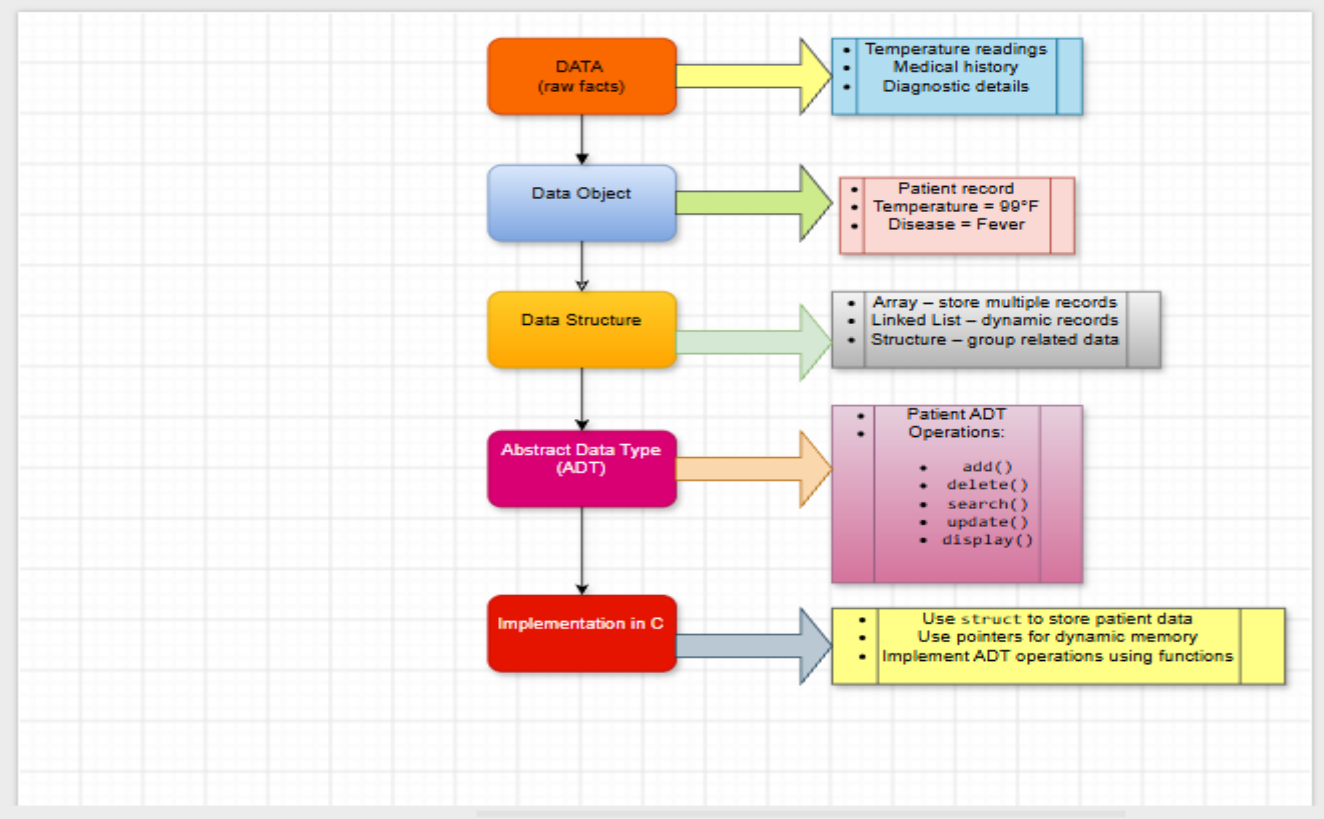
I CH.BHARATH KUMAR REDDY(192411185) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

CH.BHARATH KUMAR REDDY
Signature of the student:

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Concept Map: Data Organization in a Smart Healthcare System



1. **DATA**

Raw facts collected from a patient

- Temperature → 99°F
- Blood Pressure → 120/80 mmHg
- Heart Rate → 82 bpm
- Medical History → Diabetes
- Diagnostic Detail → Fever

These are individual pieces of raw information.

2. DATA OBJECT

Meaningful collection of related data

Patient Record

- Patient ID → 101
- Age → 45
- Temperature → 99°F
- Blood Pressure → 120/80

- Heart Rate → 82 bpm
- Medical History → Diabetes
- Diagnosis → Fever

Related data items are combined to represent one patient.

3. DATA STRUCTURE

Organizes and stores patient objects efficiently

- **Structure** → Groups different patient details
- **Array** → Stores multiple patient records
- **Linked List** → Stores dynamically changing patient records
- **Pointers** → Access patient data efficiently

Data structures make storage and access easier.

4. ABSTRACT DATA TYPE (ADT)

Defines what operations can be performed on patient data

Patient ADT

- add() → Add a new patient
- delete() → Remove a patient record
- search() → Find a patient
- update() → Modify patient details
- display() → Show patient information

ADT defines operations without focusing on implementation details.

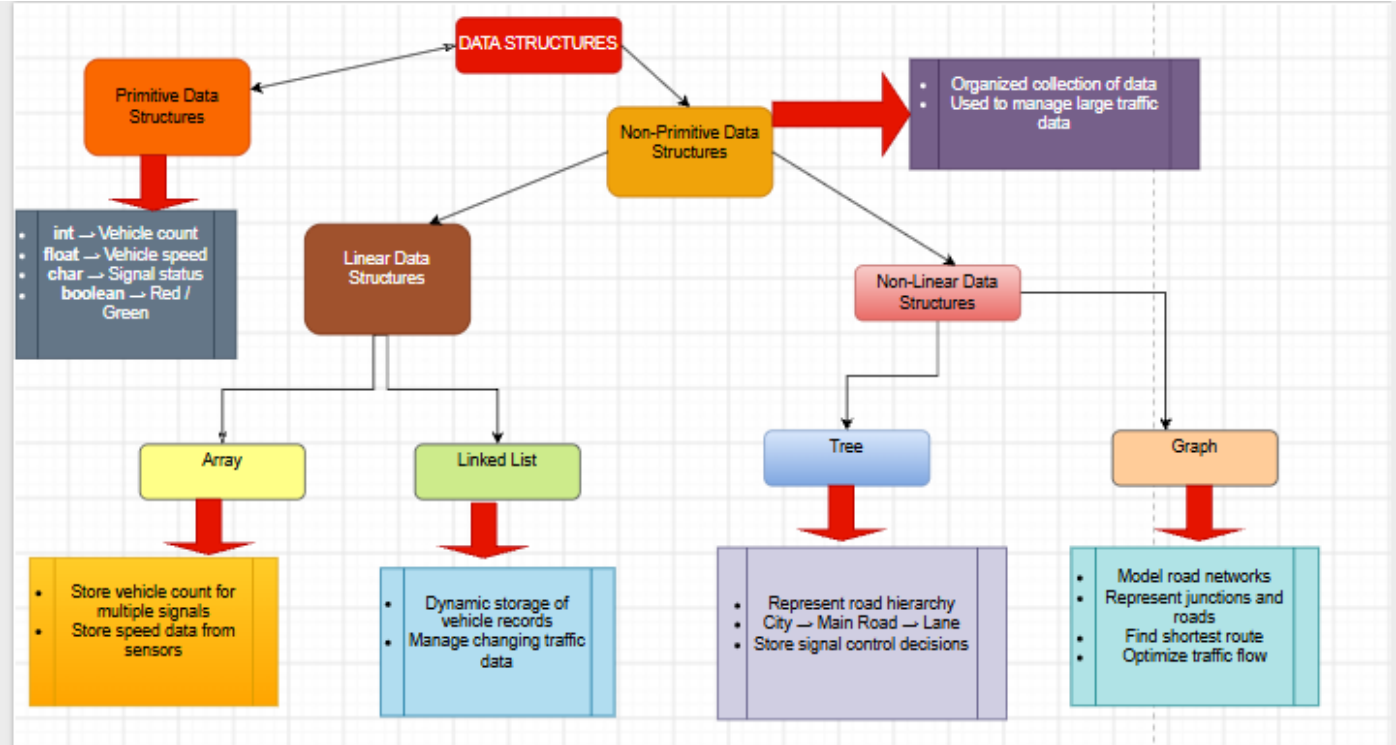
5. IMPLEMENTATION IN C

ADT is converted into actual C code

- struct Patient → Stores patient details
- **Pointers** → Access and manage memory
- **Functions** → Implement ADT operations
- malloc() → Allocate memory
- free() → Release memory

2nd ans.

Concept Map: Classification of Data Structures in Smart City Traffic Management



1. Primitive Data Structures

- **Integer (int)** → Number of vehicles
- **Float** → Vehicle speed
- **Character (char)** → Signal status
- **Boolean** → Signal ON/OFF

Basic data values used in traffic systems

2. Non-Primitive Data Structures

- Store multiple and complex data
- Used for road, vehicle and signal information

Divided into Linear and Non-Linear

3. Linear Data Structures

Array

- Vehicle count at signals
- Speed sensor data

Linked List

- Dynamic vehicle records
- Changing traffic information

Data is arranged sequentially

4. Non-Linear Data Structures

Tree

- Road hierarchy
- City → Road → Lane
- Signal control decisions

Graph

- Road network representation
- Junctions and roads
- Shortest route
- Traffic flow optimization

5. Application in Smart Traffic

- Monitor vehicle movement
- Manage traffic signals
- Represent road networks
- Find shortest routes
- Reduce traffic congestion